Ningchen MA

New Haven, CT

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EDUCATION

Yale University — **New Haven, CT**
M.S. in *Computer Science (Expected)* — ***Sep.2026 – May.2027***

McGill University — **Montreal, Canada**
B.S. in *Computer Science* **(Honours)** — ***Sep.2017-Jun.2022***

- **First Class Honours** in Computer Science | **GPA: 3.87/4.00**

WORK EXPERIENCE

Amazon — **Vancouver, Canada**
Software Development Engineer, Dept. of PXT (fulltime) — ***Oct.2022 – Jun.2026***

- **Led backend development for a production workforce planning system**, owning APIs, data modeling, testing, deployment, and operational support.
  - Designed and implemented 40+ GraphQL APIs for scenario planning, workforce calculations, and operational workflows.
  - Built hiring, attrition, and promotion workforce models backed by scalable DynamoDB services for organizational planning calculations.
- **Designed a validation algorithm that blocked invalid workforce data before it reached financial planning**, preventing duplicate and missing employee assignments from propagating into headcount and budget calculations. Replaced O(n) employee scans with O(p log p) checks over p selected entries; a 90,000-employee example needed ~3,000× fewer database reads.
- **Developed a React-based frontend**, building user-facing workflows and integrating REST APIs.
- **Built a GenAI analytics assistant** and remote MCP server end-to-end, enabling natural-language workforce queries and cutting manual data exploration from tens of minutes to seconds.

RESEARCH EXPERIENCE

Shir Group, McGill University (Research Assistant) — ***May.2022 – Sep.2022***

- Researched compiler optimization using equality saturation and IR selection; built a Swift/MLCompute toolchain for Apple silicon M1 neural accelerator.

Cloud/IoT Group, McGill University (Research Assistant) — ***Jan.2022 – Apr.2022***

- Optimized MQTT communications on ESP32 devices running RIOT OS for edge computing.

PROJECT EXPERIENCE

Torchbearer — Hack Nation 3rd Place | GitHub
Team Project, Frontend UI & Workflows — *Oct.2026*

- Built the frontend UI and workflows for Torchbearer, a voice-assisted tool that turns expert reasoning into interactive work maps and step-by-step guidance for new hires.

SKILLS

Programming: Python, Java, JavaScript, C/C++, Scala, OCaml, MIPS Assembly, Bash Script, HTML+CSS, PHP
Web / AI / APIs: React, GraphQL, REST APIs, GenAI Agents, MCP, Agentic AI-assisted Development
Amazon Web Services: Lambda, DynamoDB, AppSync, S3, CDK, CloudWatch, API Gateway, OpenSearch, ECS, Route 53, VPC